

Rocket race

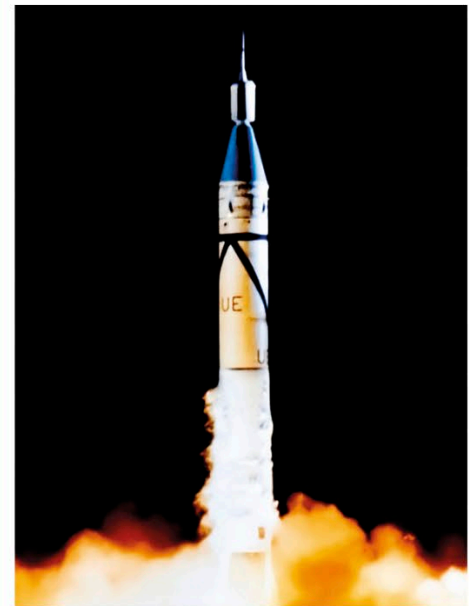
Rockets come in a wide variety of shapes and sizes, from fireworks to battlefield weapons and long-range missiles. However, the largest and most powerful rockets are those used to launch satellites and manned spacecraft into orbit and beyond. Although the first experiments with liquid-fuel rockets took place in the 1920s, it took military interest in rockets to make spaceflight a reality.

Military rockets

- **What?** V-2 rocket
- **Who?** Werner von Braun
- **Where and when?** Germany, 1942

In the 1930s, the development of rockets was mostly confined to amateur engineers. In Germany, however, a team led by Werner von Braun was ordered to develop them for the Nazi war effort. The result was the V-2 long-range missile. Its steering and guidance systems were a huge technical advance for rocketry, and in 1942, it became the first man-made object to enter space—but the V-2 was also a terrible weapon of war, responsible for an estimated 9,000 deaths by the end of World War II in 1945.

Launch of the Explorer satellite on Juno 1



Satellite launchers

- **What?** Juno I
- **Who?** Werner von Braun
- **Where and when?** US, 1958

After the Russians' launch of Sputnik 1 into orbit in 1957, American scientists were further demoralized by the explosion of their Vanguard rocket. Anxious to catch up, they turned to Werner von Braun, who was now working for the US Army. Within a month, his team launched the Explorer 1 satellite on a missile adapted into a four-stage rocket called Juno 1.